Lecture 21

Type C fundamental representations

Objectives. *A symplectic form on $V = \mathbb{C}^{2n}$ gives an $\mathfrak{sl}_2$ acting on ΛV across exterior degrees. Its primitive forms realise the fundamental modules of $\mathfrak{sp}_{2n}$.*

By the end of this lecture you will be able to:

- Compute the C_n roots, coroots and fundamental weights of $\mathfrak{sp}_{2n}$.
- Decompose $\Lambda^k V$ into primitive pieces using the Lefschetz $\mathfrak{sl}_2$.
- Prove $\Lambda^k_0 V \cong L(\omega_k)$ for $1 \le k \le n$.
- Identify $\Lambda^2_0 \mathbb{C}^4$ with the vector module of $\mathfrak{so}_5$.

Throughout, $n \ge 1$ and $V = \mathbb{C}^{2n}$. All modules are finite-dimensional over $\mathbb{C}$.

21.1 The algebra $\mathfrak{sp}_{2n}$ and its C_n data

Write the standard basis of V as $e_1, \dots, e_n, f_1, \dots, f_n$, where f_i is the $(n+i)$-th standard vector. The *symplectic form* is

$$\omega(u, v) = u^{\mathsf{T}} J v, \qquad J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}.$$

Thus

$$\omega(e_i, f_j) = \delta_{ij} = -\omega(f_j, e_i), \qquad \omega(e_i, e_j) = \omega(f_i, f_j) = 0.$$

Definition 21.1 (Symplectic Lie algebra)**.** The *symplectic Lie algebra* is

$$\mathfrak{sp}_{2n} = \{X \in \mathfrak{gl}_{2n}(\mathbb{C}) : \omega(Xu, v) + \omega(u, Xv) = 0 \text{ for all } u, v \in V\}.$$

As in Lecture 10, $\mathfrak{sp}_{2n}$ is a subalgebra. Since $\omega(Xu, v) = u^{\mathsf{T}} X^{\mathsf{T}} J v$, the defining condition is $X^{\mathsf{T}} J + J X = 0$. In $n \times n$ blocks A, B, C, D of X, it says $D = -A^{\mathsf{T}}$, $B = B^{\mathsf{T}}$ and $C = C^{\mathsf{T}}$. Hence

$$\mathfrak{sp}_{2n} = \left\{ \begin{pmatrix} A & B \\ C & -A^{\mathsf{T}} \end{pmatrix} : B = B^{\mathsf{T}},\ C = C^{\mathsf{T}} \right\}, \qquad \dim \mathfrak{sp}_{2n} = 2n^2 + n.$$

Let $\mathfrak{h}$ consist of the diagonal matrices $H(t) = \operatorname{diag}(t_1, \dots, t_n, -t_1, \dots, -t_n)$, and put $\varepsilon_i(H(t)) = t_i$. With matrix units E_{ab}, the bracket $[H(t), E_{ab}]$ is $(d_a - d_b)E_{ab}$, where d is the diagonal of $H(t)$. This gives root vectors:

$$\begin{aligned} X_{ij} &= E_{ij} - E_{n+j,n+i} \quad (i \ne j), && \text{root } \varepsilon_i - \varepsilon_j, \\ Y_{ij} &= E_{i,n+j} + E_{j,n+i} \quad (i < j), && \text{root } \varepsilon_i + \varepsilon_j, \\ Y_{ii} &= E_{i,n+i}, && \text{root } 2\varepsilon_i. \end{aligned}$$

The transposes Y_{ij}^{T} and Y_{ii}^{T} have roots $-\varepsilon_i-\varepsilon_j$ and $-2\varepsilon_i$. These $2n^2$ matrices and $\mathfrak{h}$ span $\mathfrak{sp}_{2n}$. So $\mathfrak{h}$ is its own centraliser, and we have the root decomposition of Lecture 7 with

$$R=\{\pm\varepsilon_i\pm\varepsilon_j : i<j\}\cup\{\pm2\varepsilon_i\}.$$

Take $R_+=\{\varepsilon_i-\varepsilon_j,\ \varepsilon_i+\varepsilon_j\ (i<j),\ 2\varepsilon_i\}$ and $\alpha_i=\varepsilon_i-\varepsilon_{i+1}$ for $i<n$, $\alpha_n=2\varepsilon_n$. These simple roots form the C_n base proved in Lecture 10.

For $\alpha\in R_+$ the coroot $h_\alpha\in[\mathfrak{g}_\alpha,\mathfrak{g}_{-\alpha}]$ satisfies $\alpha(h_\alpha)=2$ (Lecture 8). Matrix multiplication gives

$$\begin{aligned}h_{\varepsilon_i-\varepsilon_j}&=[X_{ij},X_{ji}]=E_{ii}-E_{jj}-E_{n+i,n+i}+E_{n+j,n+j},\\ h_{\varepsilon_i+\varepsilon_j}&=[Y_{ij},Y_{ij}^{\mathsf{T}}]=E_{ii}+E_{jj}-E_{n+i,n+i}-E_{n+j,n+j},\\ h_{2\varepsilon_i}&=[Y_{ii},Y_{ii}^{\mathsf{T}}]=E_{ii}-E_{n+i,n+i}.\end{aligned}$$

For instance,

$$X_{ij}X_{ji}=E_{ii}+E_{n+j,n+j},\qquad X_{ji}X_{ij}=E_{jj}+E_{n+i,n+i}.$$

For $\mu=\sum_i\mu_i\varepsilon_i$,

$$\mu(h_{\varepsilon_i-\varepsilon_j})=\mu_i-\mu_j,\qquad \mu(h_{\varepsilon_i+\varepsilon_j})=\mu_i+\mu_j,\qquad \mu(h_{2\varepsilon_i})=\mu_i.\tag{21.1}$$

With $h_i=h_{\alpha_i}$, (21.1) gives $\omega_k(h_i)=\delta_{ik}$ for

$$\omega_k=\varepsilon_1+\cdots+\varepsilon_k\quad(1\le k\le n),\qquad \rho=\sum_{k=1}^{n}\omega_k=\sum_{i=1}^{n}(n+1-i)\varepsilon_i.$$

The vectors e_i and f_i have weights ε_i and $-\varepsilon_i$.

Reading off columns, the root vectors satisfy

$$X_{ij}e_j=e_i\ (i<j),\qquad X_{ij}e_l=0\ (l\ne j),\qquad Y_{ij}e_l=0\ (i\le j).\tag{21.2}$$

So every positive root vector maps e_j into $\operatorname{span}\{e_i : i<j\}$.

21.2 The invariant bivector and the contraction

Let $\Lambda V=\bigoplus_{k=0}^{2n}\Lambda^kV$, with $\Lambda^jV=0$ for $j<0$. We use the exterior action and antisymmetriser in Sheet 18.2. The antisymmetriser embeds Λ^kV in $V^{\otimes k}$ as the alternating tensors. For $X\in\mathfrak{gl}(V)$, the exterior action on Λ^kV is

$$\rho_\wedge(X)(v_1\wedge\cdots\wedge v_k)=\sum_{j=1}^{k}v_1\wedge\cdots\wedge Xv_j\wedge\cdots\wedge v_k.$$

Hence $\rho_\wedge(X)$ satisfies the derivation rule

$$\rho_\wedge(X)(\xi\wedge\eta)=\rho_\wedge(X)\xi\wedge\eta+\xi\wedge\rho_\wedge(X)\eta.$$

On $\Lambda^{2n}V$, only diagonal entries of X survive in the basis wedge. Thus $\rho_\wedge(X)$ acts by their sum $\operatorname{tr}X=0$ for $X\in\mathfrak{sp}_{2n}$.

Definition 21.2 (Bivector and wedge operator). The *symplectic bivector* and the *wedge*

operator are

$$\Omega = \sum_{i=1}^{n} e_i \wedge f_i \in \Lambda^2 V, \qquad L\colon \Lambda V \to \Lambda V, \quad L(\eta) = \Omega \wedge \eta.$$

The letter L with a form as its argument is this operator. With a weight as its argument, as in $L(\omega_k)$, it is a simple module.

Lemma 21.3. *For $X \in \mathfrak{sp}_{2n}$, $\rho_\wedge(X)\Omega = 0$. Hence L commutes with every $\rho_\wedge(X)$.*

Proof. The columns of X are

$$Xe_j = \sum_i (A_{ij}e_i + C_{ij}f_i), \qquad Xf_j = \sum_i (B_{ij}e_i - A_{ji}f_i).$$

Therefore

$$\begin{aligned}\rho_\wedge(X)\Omega &= \sum_j (Xe_j \wedge f_j + e_j \wedge Xf_j) \\ &= \sum_{i,j} \big(A_{ij}\, e_i \wedge f_j - A_{ji}\, e_j \wedge f_i\big) + \sum_{i,j} C_{ij}\, f_i \wedge f_j + \sum_{i,j} B_{ij}\, e_j \wedge e_i.\end{aligned}$$

The A-terms cancel after exchanging i and j in the second term. The C- and B-sums vanish, since symmetric coefficients meet antisymmetric wedges.

The derivation rule now gives $\rho_\wedge(X)(\Omega \wedge \eta) = \Omega \wedge \rho_\wedge(X)\eta$. ∎

The invariant bivector shows why a further operator is needed. In type A, each $\Lambda^k \mathbb{C}^m$ with $1 \le k \le m-1$ is a simple $\mathfrak{sl}_m$-module (Exercise 18.2). For $\mathfrak{sp}_{2n}$ with $n \ge 2$ this fails, since $\Lambda^2 V$ contains the invariant line $\mathbb{C}\Omega$.

For example, $\dim \Lambda^2 \mathbb{C}^4 = 6$. The contraction we now define is an intertwiner $\iota_\omega\colon \Lambda^2\mathbb{C}^4 \to \mathbb{C}$ with $\iota_\omega \Omega = 2$. So

$$\Lambda^2 \mathbb{C}^4 = \mathbb{C}\Omega \oplus \ker \iota_\omega, \qquad 6 = 1 + 5.$$

This is a preview. Section 21.5 carries out the calculation, and Theorem 21.8 proves that the five-dimensional kernel is simple. No proof below uses this example.

For $\varphi \in V^*$, the *interior product* is

$$\iota_\varphi(v_1 \wedge \cdots \wedge v_k) = \sum_{j=1}^{k} (-1)^{j-1} \varphi(v_j)\, v_1 \wedge \cdots \wedge \widehat{v_j} \wedge \cdots \wedge v_k.$$

The right side is alternating and multilinear, so ι_φ is well defined. For $v \in V$, let $m_v\colon \Lambda V \to \Lambda V$ be given by $m_v(\eta) = v \wedge \eta$. Then

$$m_u m_v + m_v m_u = 0, \qquad \iota_\varphi \iota_\psi + \iota_\psi \iota_\varphi = 0, \qquad \iota_\varphi m_v + m_v \iota_\varphi = \varphi(v)\,\mathrm{id}_{\Lambda V}. \tag{21.3}$$

The third identity is the defining formula with $v_1 = v$. For the second, in $\iota_\varphi \iota_\varphi$ the terms removing v_p then v_q, and v_q then v_p, have opposite signs. So $\iota_\varphi^2 = 0$, and expanding $\iota_{\varphi+\psi}^2 = 0$ gives anticommutation.

Let (e_i^*, f_i^*) be the dual basis. As a bilinear form,

$$\omega(u, v) = \sum_i \big(e_i^*(u) f_i^*(v) - f_i^*(u) e_i^*(v)\big).$$

Definition 21.4 (Contraction and primitive forms)**.** The *contraction* is the map

$$\iota_\omega = \sum_{i=1}^{n} \iota_{f_i^*}\iota_{e_i^*} \colon \Lambda^k V \to \Lambda^{k-2} V.$$

The space of *primitive k-forms* is

$$\Lambda_0^k V = \ker\bigl(\iota_\omega \colon \Lambda^k V \to \Lambda^{k-2} V\bigr).$$

For $u, v \in V$,

$$\begin{aligned}\iota_\omega(u \wedge v) &= \sum_i \iota_{f_i^*}\bigl(e_i^*(u)v - e_i^*(v)u\bigr)\\ &= \sum_i \bigl(e_i^*(u)f_i^*(v) - e_i^*(v)f_i^*(u)\bigr) = \omega(u,v).\end{aligned}$$

Lemma 21.5. *The contraction ι_ω commutes with every $\rho_\wedge(X)$, $X \in \mathfrak{sp}_{2n}$. Hence each $\Lambda_0^k V$ is a submodule.*

Proof. First, for $\varphi \in V^*$,

$$\iota_\varphi\, \rho_\wedge(X) = \rho_\wedge(X)\, \iota_\varphi + \iota_{\varphi \circ X}. \tag{21.4}$$

Apply the left side to $v_1 \wedge \cdots \wedge v_k$. This gives the sum over l of $\iota_\varphi(v_1 \wedge \cdots \wedge Xv_l \wedge \cdots \wedge v_k)$. Removing position l gives $(-1)^{l-1}\varphi(Xv_l)$ times the monomial without v_l. These terms sum to $\iota_{\varphi\circ X}$. Removing a position $j \neq l$ gives $(-1)^{j-1}\varphi(v_j)$ times the monomial without v_j and with Xv_l in place of v_l. Summed over l and $j \neq l$, these are the terms of $\rho_\wedge(X)\iota_\varphi(v_1 \wedge \cdots \wedge v_k)$.

Second, rows i and $n+i$ of X give

$$e_i^* \circ X = \sum_j (A_{ij}e_j^* + B_{ij}f_j^*), \qquad f_i^* \circ X = \sum_j (C_{ij}e_j^* - A_{ji}f_j^*).$$

Using (21.4) twice,

$$\begin{aligned}[\iota_\omega, \rho_\wedge(X)] &= \sum_i \bigl(\iota_{f_i^*\circ X}\iota_{e_i^*} + \iota_{f_i^*}\iota_{e_i^*\circ X}\bigr)\\ &= \sum_{i,j} \bigl(C_{ij}\iota_{e_j^*}\iota_{e_i^*} - A_{ji}\iota_{f_j^*}\iota_{e_i^*} + A_{ij}\iota_{f_i^*}\iota_{e_j^*} + B_{ij}\iota_{f_i^*}\iota_{f_j^*}\bigr).\end{aligned}$$

The A-terms cancel after exchanging i and j. The C- and B-terms vanish by symmetry and (21.3). ∎

21.3 The Lefschetz $\mathfrak{sl}_2$ and primitive decomposition

The wedge operator L raises the exterior degree by 2, and the contraction ι_ω lowers it by 2. So neither operator acts on a single $\Lambda^k V$. Together with the operator H below, they span an auxiliary $\mathfrak{sl}_2$. This $\mathfrak{sl}_2$ acts on all of ΛV, across exterior degrees. It is not a subalgebra of $\mathfrak{sp}_{2n}$, whose elements preserve each degree.

Let H act on $\Lambda^k V$ as multiplication by $n - k$.

Proposition 21.6 (Lefschetz relation)**.** *On ΛV,*

$$[\iota_\omega, L] = H, \qquad [H, \iota_\omega] = 2\iota_\omega, \qquad [H, L] = -2L.$$

Hence $e \mapsto \iota_\omega$, $h \mapsto H$, $f \mapsto L$ is a representation of $\mathfrak{sl}_2$ on ΛV. These three operators commute with every $\rho_\wedge(X)$, $X \in \mathfrak{sp}_{2n}$.

Proof. For $\eta \in \Lambda^k V$,

$$[H, L]\eta = (n-k-2)L\eta - (n-k)L\eta = -2L\eta.$$

The same calculation, with degree $k-2$, gives $[H, \iota_\omega] = 2\iota_\omega$.

For the first relation, put $a_i = m_{e_i}$, $b_i = m_{f_i}$, $a_i^* = \iota_{e_i^*}$ and $b_i^* = \iota_{f_i^*}$. Then

$$L = \sum_i a_i b_i, \qquad \iota_\omega = \sum_i b_i^* a_i^*.$$

By (21.3), any two of these operators anticommute, except

$$a_i^* a_i + a_i a_i^* = 1, \qquad b_i^* b_i + b_i b_i^* = 1.$$

If p and q each anticommute with r and s, then pq commutes with rs. Hence $b_j^* a_j^*$ commutes with $a_i b_i$ for $j \neq i$, and

$$[\iota_\omega, L] = \sum_{i=1}^{n} [b_i^* a_i^*, a_i b_i].$$

Fix i and drop the index. Put $N_a = aa^*$ and $N_b = bb^*$. Then N_a commutes with b, b^*, and N_b commutes with a, a^*. So

$$\begin{aligned} b^* a^* a b &= b^*(1 - N_a)b = (1 - N_a) b^* b = (1 - N_a)(1 - N_b), \\ ab\, b^* a^* &= a N_b a^* = N_b\, aa^* = N_a N_b. \end{aligned}$$

Subtracting, $[b_i^* a_i^*, a_i b_i] = 1 - N_{a_i} - N_{b_i}$.

On a wedge of distinct basis vectors, N_{a_i} is 1 if e_i is a factor and 0 otherwise, because $\iota_{e_i^*}$ removes e_i with a sign and m_{e_i} restores it with the same sign. The same holds for N_{b_i} and f_i. Thus $\sum_i (N_{a_i} + N_{b_i}) = k$ on $\Lambda^k V$, and

$$[\iota_\omega, L] = \sum_{i=1}^{n} (1 - N_{a_i} - N_{b_i}) = n - k = H \quad \text{on } \Lambda^k V.$$

Commutation with $\rho_\wedge(X)$ follows from Lemmas 21.3 and 21.5, since H is scalar on each submodule $\Lambda^k V$. ∎

By Lecture 6, a finite-dimensional $\mathfrak{sl}_2$-module is a direct sum of irreducibles V_m. Each V_m has a basis $v_j = f^j v_0$, $0 \le j \le m$, with $hv_j = (m-2j)v_j$ and $fv_m = 0$. Moreover $\ker e \cap V_m = \mathbb{C}v_0$, of h-weight $m \ge 0$. Also $ev_j = j(m - j + 1)v_{j-1}$ for $1 \le j \le m$.

Figure 21.1 previews the decomposition for $n = 2$. A nonzero primitive vector η of degree r is killed by $e = \iota_\omega$ and has h-weight $n - r$. So η starts a string $\eta, L\eta, L^2\eta, \ldots$ in the degrees $r, r+2, r+4, \ldots$. With $m = n - r$, the formula for e gives

$$\iota_\omega L^j \eta = j(n - r - j + 1)\, L^{j-1}\eta.$$

The theorem below proves that such strings fill ΛV, and it proves the dimensions shown in the figure.

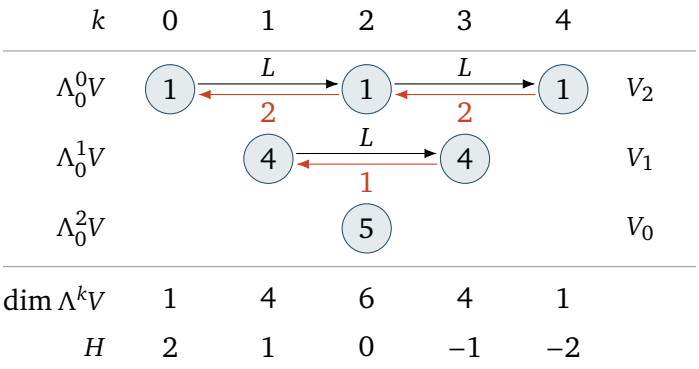


Figure 21.1: Lefschetz strings for $n = 2$: a circle in row $\Lambda_0^r V$ and column $k = r + 2j$ gives $\dim L^j \Lambda_0^r V$. Black arrows are L, coloured arrows are ι_ω with the scalar $j(n - r - j + 1)$, and each column sums to $\dim \Lambda^k V$.

Theorem 21.7 (Primitive decomposition)**.**

(a) $\Lambda_0^k V = 0$ *for* $k > n$.

(b) If $0 \le r \le n$ *and* $0 \ne \eta \in \Lambda_0^r V$*, then* $L^j \eta \ne 0$ *for* $0 \le j \le n - r$*, and* $L^{n-r+1}\eta = 0$.

(c) For $0 \le k \le 2n$*,*

$$\Lambda^k V = \bigoplus_{j=\max(0,k-n)}^{\lfloor k/2 \rfloor} L^j\big(\Lambda_0^{k-2j} V\big),$$

and each $L^j \colon \Lambda_0^{k-2j} V \to L^j \Lambda_0^{k-2j} V$ *is an isomorphism of* $\mathfrak{sp}_{2n}$*-modules.*

(d) For $0 \le k \le n$*, the map* $\iota_\omega \colon \Lambda^k V \to \Lambda^{k-2} V$ *is onto, and*

$$\dim \Lambda_0^k V = \binom{2n}{k} - \binom{2n}{k-2},$$

with $\binom{2n}{-1} = \binom{2n}{-2} = 0$.

Proof. Decompose $\Lambda V = \bigoplus_s U_s$ into irreducible $\mathfrak{sl}_2$-submodules. Let $U_s \cong V_{m_s}$ have basis $v_{s,j} = L^j v_{s,0}$. The eigenvalues $n - k$ of H are distinct, so the H-eigenspaces are exactly the $\Lambda^k V$. Hence $v_{s,0} \in \Lambda^{r_s} V$ with $m_s = n - r_s$, and $v_{s,j} \in \Lambda^{r_s+2j} V$.

Since ι_ω preserves each U_s, its kernel is $\bigoplus_s \mathbb{C} v_{s,0}$. Taking degree r, the vectors $v_{s,0}$ with $r_s = r$ form a basis of $\Lambda_0^r V$. As $m_s \ge 0$, every $r_s \le n$. This proves (a).

For (b), write $\eta = \sum c_s v_{s,0}$ over $r_s = r$, so all $m_s = n - r$. Then $L^j \eta = \sum c_s v_{s,j}$. It is nonzero for $j \le n - r$ by independence, and zero for $j = n - r + 1$.

For (c), the homogeneous basis vectors $v_{s,j}$ with $r_s + 2j = k$ form a basis of $\Lambda^k V$. For fixed j they are the images under L^j of the basis of $\Lambda_0^{k-2j} V$, and they exist exactly when $j \le n - (k - 2j)$, that is, $j \ge k - n$. By (b), L^j is injective on $\Lambda_0^{k-2j} V$ in this range. It commutes with $\mathfrak{sp}_{2n}$ by Proposition 21.6.

For (d), compare (c) in degrees k and $k - 2$ when $2 \le k \le n$:

$$\dim \Lambda^k V = \dim \Lambda_0^k V + \sum_{j \ge 1} \dim \Lambda_0^{k-2j} V = \dim \Lambda_0^k V + \dim \Lambda^{k-2} V.$$

Rank and nullity show that ι_ω is onto. For $k = 0, 1$, its target is zero. ∎

Figure 21.2 shows the decomposition for $n = 3$. A string starting in degree r ends in degree $2n - r$. This is the range of j in the theorem. For example, in degree $k = 4$ the

summands have $j = 1$ and $j = 2$, from $\Lambda_0^2 V$ and $\Lambda_0^0 V$, and $15 = 14 + 1$.

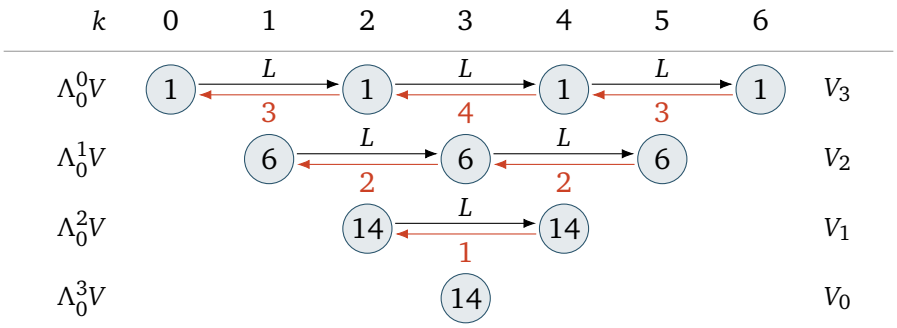


Figure 21.2: Lefschetz strings for $n = 3$, read as in Figure 21.1. A circle gives a multiplicity, not one string: for example, the row $\Lambda_0^2 V$ is 14 copies of V_1.

21.4 Primitive exterior powers are fundamental

The method of this section has two steps. A highest vector gives a simple submodule. We then compare its Weyl dimension with the dimension of the proposed ambient module. When the two dimensions agree, the submodule is the whole module. Lectures 22 and 23 use the same method.

Theorem 21.8. *For $1 \le k \le n$,*

$$\Lambda_0^k V \cong L(\omega_k), \qquad \dim L(\omega_k) = \binom{2n}{k} - \binom{2n}{k-2} = \frac{2(n+1-k)\,(2n+1)!}{k!\,(2n+2-k)!}.$$

Proof. Step 1: a highest vector. Since $k \le n$, the vector $v_k = e_1 \wedge \cdots \wedge e_k$ is defined. Each $\iota_{e_i^*} v_k$ is a wedge of e's, so $\iota_{f_i^*}$ kills it. Thus $v_k \in \Lambda_0^k V$.

$$\rho_\wedge(H(t))v_k = (t_1 + \cdots + t_k)v_k,$$

so v_k has weight ω_k. Let X be a positive root vector. By (21.2), Xe_j lies in the span of e_i with $i < j \le k$. Each such e_i is already a factor, so every term of $\rho_\wedge(X)v_k$ vanishes.

Step 2: the generated submodule. Let W be the submodule generated by v_k. By PBW (Lecture 11), $W = U(\mathfrak{n}_-)v_k$. Monomials in negative root vectors lower weights, so $W_{\omega_k} = \mathbb{C}v_k$. The cyclic-module lemma of Lecture 13 applies to this finite-dimensional weight module. Hence $W \cong L(\omega_k)$.

Step 3: the Weyl dimension. By Lecture 17 and (21.1),

$$\dim L(\lambda) = \frac{F(\lambda+\rho)}{F(\rho)}, \qquad F(\mu) = \prod_{\alpha \in R_+} \mu(h_\alpha) = \prod_{i<j} (\mu_i^2 - \mu_j^2) \prod_i \mu_i.$$

For a finite set S of positive integers put $F_S = \prod_{a \in S} a \prod_{a>b \in S} (a^2 - b^2)$. The coordinates of ρ form $\{1, \dots, n\}$. Those of $\omega_k + \rho$ are $n + 2 - i$ for $i \le k$ and $n + 1 - i$ for $i > k$. They form $S = T \setminus \{m\}$, where $T = \{1, \dots, n+1\}$ and $m = n + 1 - k$. Both coordinate lists are strictly

decreasing, so $F(\rho) = F_{\{1,\dots,n\}}$ and $F(\omega_k + \rho) = F_S$. Both sets are T with one integer removed. So we expand F_T in two ways:

$$F_T = F_{\{1,\dots,n\}} \cdot (n+1) \prod_{b=1}^{n} \big((n+1)^2 - b^2\big),$$
$$F_T = F_S \cdot m \prod_{b \in T,\, b \neq m} |m - b|\,(m+b).$$

The first product is

$$(n+1) \cdot n! \cdot \frac{(2n+1)!}{(n+1)!} = (2n+1)!.$$

In the second,

$$\prod_{b \neq m} |m-b| = (m-1)!\,(n+1-m)!, \qquad \prod_{b \neq m} (m+b) = \frac{(n+1+m)!}{m! \cdot 2m}.$$

Equating the two expressions for F_T,

$$\dim L(\omega_k) = \frac{F_S}{F_{\{1,\dots,n\}}} = \frac{2m\,(2n+1)!}{(n+1-m)!\,(n+1+m)!} = \frac{2(n+1-k)\,(2n+1)!}{k!\,(2n+2-k)!}.$$

Step 4: comparison. Over the common denominator $k!(2n+2-k)!$,

$$\binom{2n}{k} - \binom{2n}{k-2} = \frac{(2n)!\,\big[(2n+2-k)(2n+1-k) - k(k-1)\big]}{k!\,(2n+2-k)!}.$$

With $N = 2n+1$, the bracket is

$$(N+1-k)(N-k) - k(k-1) = N(N+1-2k) = 2(2n+1)(n+1-k).$$

For $k = 1$, the convention $\binom{2n}{-1} = 0$ agrees with $k(k-1) = 0$. So $\dim W = \dim \Lambda^k_0 V$ by Theorem 21.7(d). Since $W \subseteq \Lambda^k_0 V$, the two are equal. ∎

Theorems 21.7(c) and 21.8, with $\Lambda^0_0 V = \mathbb{C}$, give for $0 \le k \le n$

$$\Lambda^k V \cong \bigoplus_{j=0}^{\lfloor k/2 \rfloor} L(\omega_{k-2j}), \qquad \omega_0 = 0.$$

21.5 The case of $\mathfrak{sp}_4$

Example 21.9 ($\Lambda^2\mathbb{C}^4 = \mathbb{C} \oplus L(\omega_2)$)**.** Let $n = 2$. Order the basis of $\Lambda^2 V$ as

$$e_1 \wedge e_2,\ e_1 \wedge f_1,\ e_1 \wedge f_2,\ e_2 \wedge f_1,\ e_2 \wedge f_2,\ f_1 \wedge f_2.$$

Since $\iota_\omega(u \wedge v) = \omega(u, v)$, the matrix of $\iota_\omega \colon \Lambda^2 V \to \mathbb{C}$, using the basis 1 of $\mathbb{C}$, is

$$D(\iota_\omega) = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Its kernel $\Lambda^2_0 V$ has basis and weights

vector	$e_1 \wedge e_2$	$e_1 \wedge f_2$	$e_1 \wedge f_1 - e_2 \wedge f_2$	$e_2 \wedge f_1$	$f_1 \wedge f_2$
weight	$\varepsilon_1 + \varepsilon_2$	$\varepsilon_1 - \varepsilon_2$	0	$-\varepsilon_1 + \varepsilon_2$	$-\varepsilon_1 - \varepsilon_2$

Also $\iota_\omega \Omega = 2 \neq 0$. Hence

$$\Lambda^2 V = \mathbb{C}\Omega \oplus \Lambda^2_0 V \cong \mathbb{C} \oplus L(\omega_2), \qquad 6 = 1 + 5.$$

Now compare with $\mathfrak{so}_5$. Let vol $= e_1 \wedge f_1 \wedge e_2 \wedge f_2$. Define the bilinear form $\beta\colon \Lambda^2 V \times \Lambda^2 V \to \mathbb{C}$ by

$$x \wedge y = \beta(x, y)\,\mathrm{vol}.$$

It is symmetric, because 2-forms commute. Write Xx for $\rho_\wedge(X)x$. Since $\rho_\wedge(X) = 0$ on $\Lambda^4 V$, the form β is invariant:

$$0 = \rho_\wedge(X)(x \wedge y) = Xx \wedge y + x \wedge Xy = \big(\beta(Xx, y) + \beta(x, Xy)\big)\mathrm{vol}.$$

Weights add under $\wedge$ and vol has weight 0. So only pairs of opposite weights can pair nontrivially. Reordering wedges gives

$$\begin{aligned} e_1 \wedge e_2 \wedge f_1 \wedge f_2 &= -\mathrm{vol}, \\ e_1 \wedge f_2 \wedge e_2 \wedge f_1 &= -\mathrm{vol}, \\ (e_1 \wedge f_1 - e_2 \wedge f_2) \wedge (e_1 \wedge f_1 - e_2 \wedge f_2) &= -2\,\mathrm{vol}. \end{aligned}$$

Hence β is nondegenerate on $\Lambda^2_0 V$.

The action therefore gives a homomorphism $\mathfrak{sp}_4 \to \mathfrak{so}(\Lambda^2_0 V, \beta)$. Its kernel K is an ideal, stable under $\operatorname{ad}\mathfrak{h}$. So K is the sum of $K \cap \mathfrak{h}$ and the root spaces it contains. The weights $\varepsilon_1 \pm \varepsilon_2$ separate $\mathfrak{h}$, so $K \cap \mathfrak{h} = 0$. If $\mathfrak{g}_\alpha \subseteq K$, then $h_\alpha \in [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subseteq K$, which is impossible. Thus $K = 0$. All nondegenerate symmetric forms on $\mathbb{C}^5$ are equivalent, so $\mathfrak{so}(\Lambda^2_0 V, \beta) \cong \mathfrak{so}_5$. Moreover,

$$\dim \mathfrak{sp}_4 = 10 = \dim \mathfrak{so}_5.$$

Hence $\mathfrak{sp}_4 \cong \mathfrak{so}_5$, and $\Lambda^2_0 \mathbb{C}^4$ becomes the vector module $\mathbb{C}^5$.

In the B_2 coordinates of Lecture 10, put

$$\varepsilon'_1 = \varepsilon_1 + \varepsilon_2, \qquad \varepsilon'_2 = \varepsilon_1 - \varepsilon_2.$$

The weights of $\Lambda^2_0 V$ are then $0, \pm\varepsilon'_1, \pm\varepsilon'_2$, as for the vector module. The B_2 simple roots are

$$\begin{aligned} \varepsilon'_1 - \varepsilon'_2 &= 2\varepsilon_2 = \alpha_2, \\ \varepsilon'_2 &= \varepsilon_1 - \varepsilon_2 = \alpha_1. \end{aligned}$$

Thus the labels are exchanged. The module $L(\omega_2)$ for C_2 is $L(\omega'_1)$ for B_2.

In the same way, $\varepsilon_1 = \frac{1}{2}(\varepsilon'_1 + \varepsilon'_2)$ is the fundamental weight ω'_2 of B_2. So the defining module $V = L(\omega_1)$ is $L(\omega'_2)$ for B_2. A weight $x\varepsilon_1 + y\varepsilon_2$ has the coordinates $\big(\frac{1}{2}(x+y), \frac{1}{2}(x-y)\big)$ with respect to $\varepsilon'_1, \varepsilon'_2$. Hence the weights of V become $(\pm\frac{1}{2}, \pm\frac{1}{2})$. Lecture 24 and Exercise 24.2 construct this B_2 module by another method.

Figure 21.3 shows the two coordinate systems together.

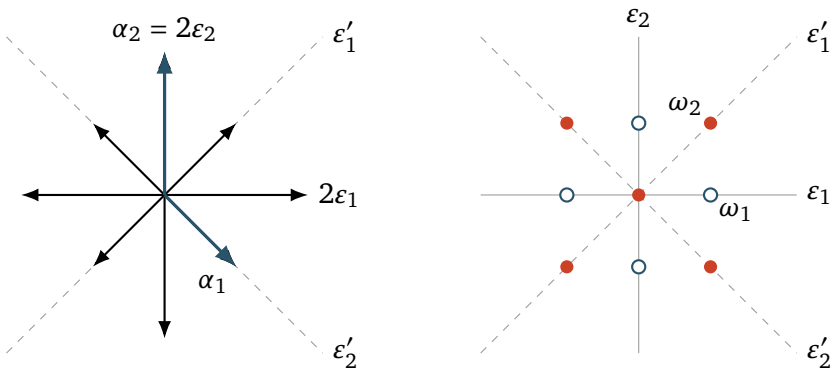


Figure 21.3: The roots of $\mathfrak{sp}_4 \cong \mathfrak{so}_5$ and the weights of two modules, with $\varepsilon_1, \varepsilon_2$ as unit axes and the ε'-directions dashed. The coloured arrows are the simple roots; filled points are the weights of $\Lambda^2_0 V$, and open points are the weights of V.

Self-checks

(1) Compute $\dim \Lambda^3_0 \mathbb{C}^8$ for $\mathfrak{sp}_8$.
(2) Why is $\Lambda^{n+1}_0 V = 0$, without any dimension count?
(3) Use Theorem 21.7(c) to prove that $\Lambda^4 V = \mathbb{C}\,\Omega \wedge \Omega$ for $\mathfrak{sp}_4$.

Answers. (1) $\binom{8}{3} - \binom{8}{1} = 56 - 8 = 48$. (2) Such a vector is killed by $e = \iota_\omega$ and has h-weight -1. A nonzero highest vector of an $\mathfrak{sl}_2$-module has weight ≥ 0 (Lecture 6). (3) Here $k = 4$, so $j \geq k - n = 2$. Only $L^2 \Lambda^0_0 V$ occurs: $\Lambda^4 V = \mathbb{C}\,\Omega \wedge \Omega$.

Summary

The operators L, ι_ω and H span an $\mathfrak{sl}_2$ acting across exterior degrees and commuting with $\mathfrak{sp}_{2n}$. Primitive forms vanish above degree n, and $\Lambda^k_0 V \cong L(\omega_k)$ for $1 \leq k \leq n$.

Reading

William Fulton and Joe Harris, *Representation Theory: A First Course*, Lectures 16–17. Roger Carter, *Lie Algebras of Finite and Affine Type*, Sections 13.6–13.7. Exercise Sheet 21 gives further calculations.